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Class: AIDS

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Sub : Python For Data Science

Aim 9- Building a Simple Data Science Pipeline: Integrate all learned techniques in previous labs into a complete workflow.

Theory :

1. What is a Data Science Pipeline?
   A **Data Science pipeline** is a structured workflow that automates the process of collecting, cleaning, analyzing, and modeling data to generate insights. It ensures consistency, reproducibility, and efficiency in handling data projects.

Key Stages with Examples:

- **Data Collection** → Gathering raw data (e.g., customer purchase records, sensor readings).
- **Data Cleaning** → Handling missing values, duplicates, or outliers (e.g., replacing nulls with mean values).
- **Exploratory Data Analysis (EDA)** → Visualizing distributions, correlations (e.g., plotting histograms, scatter plots).
- **Feature Engineering** → Creating new variables (e.g., extracting "day of week" from a timestamp).
- **Feature Scaling** → Normalizing or standardizing values (e.g., scaling income values for regression).
- **Model Building** → Training ML models (e.g., decision trees, logistic regression).
- **Evaluation** → Measuring accuracy, precision, recall, etc.
- **Deployment** → Integrating the model into applications (e.g., recommendation systems).

2. What is Exploratory Data Analysis (EDA)?

EDA is the process of **summarizing and visualizing data** to uncover patterns, detect anomalies, and test hypotheses before applying models.

Types of Analysis:

- **Univariate Analysis** → Examines one variable at a time.
  - Example: Histogram of customer ages shows skewness toward younger buyers.
- **Bivariate Analysis** → Examines relationships between two variables.
  - Example: Scatter plot of "advertising spend vs. sales" shows positive correlation.

These analyses help in understanding distributions, relationships, and guiding feature selection.

3. Label Encoding vs OneHot Encoding

Both are techniques to convert categorical data into numerical form for ML models.

Aspect	Label Encoding	OneHot Encoding
Definition	Assigns each category a unique integer	Creates binary columns for each category
Example	Colors: Red=0, Blue=1, Green=2	Colors: Red=[1,0,0], Blue=[0,1,0], Green=[0,0,1]
When to Use	For ordinal data (categories with order, e.g., "Low, Medium, High")	For nominal data (no order, e.g., "Dog, Cat, Bird")
Risk	May mislead models into thinking categories have numeric relationships	Increases dimensionality (many columns)

4. Why is Feature Scaling Required?

Feature scaling ensures that variables with different ranges (e.g., income in lakhs vs. age in years) do not dominate model training. Many algorithms (like KNN, SVM, gradient descent) are sensitive to scale.

Comparison:

- **Normalization (Min-Max Scaling)**
  - Scales values to range [0,1].
  - Formula: (\frac{x - \text{min}}{\text{max} - \text{min}})
  - Best for bounded data (e.g., percentages).

- **Standardization (Z-score Scaling)**
  - Centers data around mean 0 with standard deviation 1.
  - Formula: (\frac{x - \mu}{\sigma})
  - Best when data follows Gaussian distribution or when outliers exist.

5. What is Feature Engineering?

Feature engineering is the process of **creating new variables** or transforming existing ones to improve model performance.

Examples:

- **Date Features** → Extracting "month" or "day of week" from timestamps.
- **Text Features** → Creating word counts or sentiment scores from reviews.
- **Interaction Features** → Combining variables (e.g., "price × quantity" = revenue).
- **Domain-Specific Features** → In healthcare, BMI = weight/height^2.

By capturing hidden patterns, engineered features often make models more accurate and interpretable.

```
from google.colab import files

uploaded = files.upload()

for fn in uploaded.keys():
  print(f'User uploaded file "{fn}" with length {len(uploaded[fn])} bytes')
```

Choose Files employee_...ce_data.csv
employee_performance_data.csv(text/csv) - 1372 bytes, last modified: 4/5/2026 - 100% done

```
Saving employee_performance_data.csv to employee_performance_data (1).csv
User uploaded file "employee_performance_data (1).csv" with length 1372 bytes
```

```
# Step 1: Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, OneHotEncoder, StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Step 2: Load Dataset
# Example dataset (you can replace with your own CSV)
df = pd.read_csv("employee_performance_data (1).csv")


print("First 5 rows:")
print(df.head())

# Step 3: Data Cleaning
# Handling missing values
df.fillna(df.mean(numeric_only=True), inplace=True)

# Drop duplicates
df.drop_duplicates(inplace=True)

# Step 4: Exploratory Data Analysis (EDA)

# Univariate Analysis
df.hist(figsize=(10,8))
plt.show()

# Bivariate Analysis
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.show()

# Step 5: Feature Engineering
# Example: Create new feature
if 'Age' in df.columns:
    df['Age_Group'] = pd.cut(df['Age'], bins=[0,18,35,60,100], labels=['Teen','Young','Adult','Senior'])

# Step 6: Encoding Categorical Variables
label_encoder = LabelEncoder()
```

```
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# Include 'object' and 'category' dtypes for encoding
for col in df.select_dtypes(include=['object', 'category']).columns:
    df[col] = label_encoder.fit_transform(df[col])

# Step 7: Define Features and Target
X = df.drop('Performance_Rating', axis=1)   # Replace 'target' with your column
y = df['Performance_Rating']

# Step 8: Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Step 9: Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Step 10: Model Training
model = LinearRegression()
model.fit(X_train, y_train)

# Step 11: Prediction
y_pred = model.predict(X_test)

# Step 12: Evaluation
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Mean Squared Error:", mse)
print("R2 Score:", r2)
```
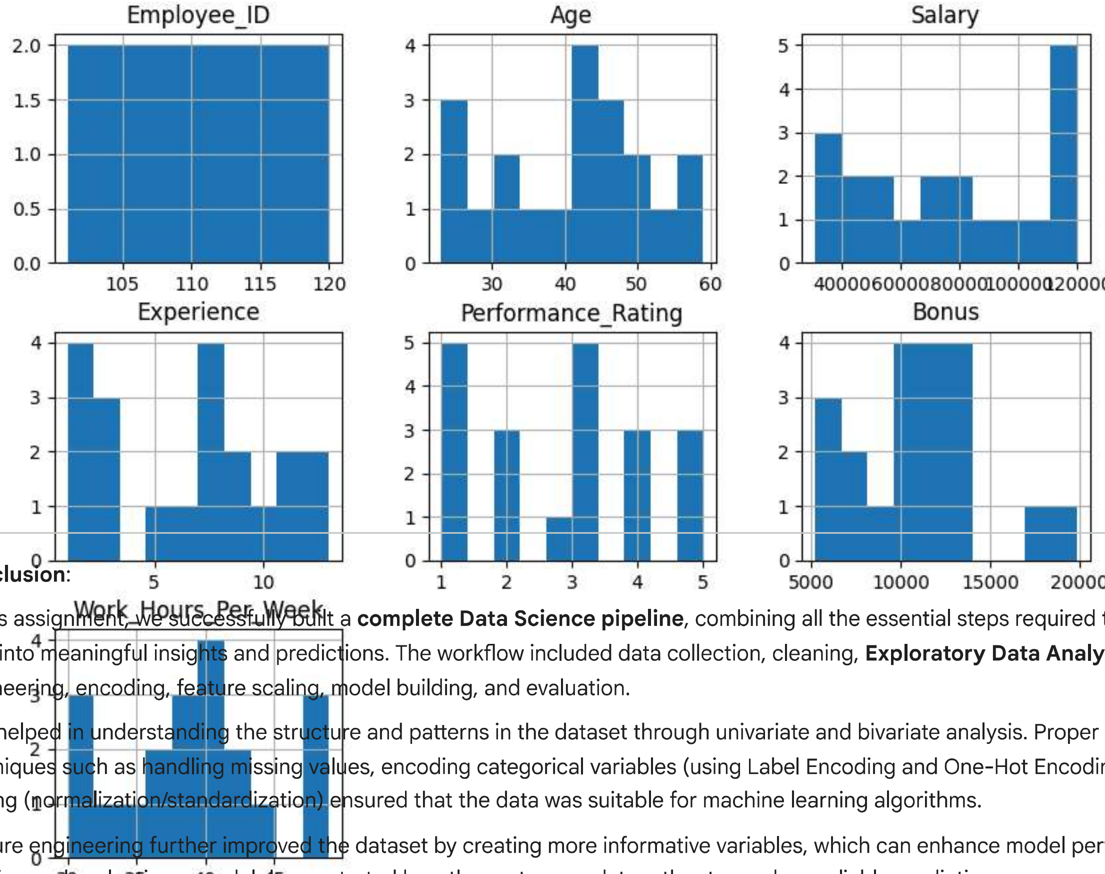

```
First 5 rows:
   Employee_ID     Name  Age  Gender Department   Join_Date    Salary  \
0          101    Alice   50  Female  Marketing  2009-01-18   62606.0
1          102      Bob   36  Female  Marketing  2012-01-01   41534.0
2          103  Charlie   29  Female         IT  2007-02-15       NaN
3          104    David   42    Male  Marketing  2015-12-29   31016.0
4          105      Eve   40  Female         IT  2005-02-04  119789.0

   Experience  Performance_Rating    Bonus  Work_Hours_Per_Week
0        13.0                 2.0   9488.0                 44.0
1         1.0                 1.0   5206.0                 40.0
2         9.0                 4.0  10134.0                 33.0
3         7.0                 1.0  10977.0                 42.0
4         9.0                 4.0  12721.0                 36.0
```



Conclusion:

In this assignment, we successfully built a **complete Data Science pipeline**, combining all the essential steps required to transform raw data into meaningful insights and predictions. The workflow included data collection, cleaning, **Exploratory Data Analysis (EDA)**, feature engineering, encoding, feature scaling, model building, and evaluation.

EDA helped in understanding the structure and patterns in the dataset through univariate and bivariate analysis. Proper preprocessing techniques such as handling missing values, encoding categorical variables (using Label Encoding and One-Hot Encoding), and feature scaling (normalization/standardization) ensured that the data was suitable for machine learning algorithms.

Feature engineering further improved the dataset by creating more informative variables, which can enhance model performance. Finally, building and evaluating a model demonstrated how these steps work together to produce reliable predictions.

Overall, this pipeline provides a structured and efficient approach to solving real-world data problems and forms the foundation for developing advanced machine learning solutions.